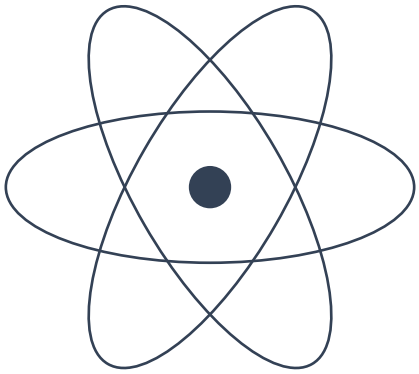
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Algebra I

Detailed Solutions Manual

Olympiad Introductory Training • Math

SPhOH Team

FIRST EDITION • 2026

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CHAPTER 1

Algebra I Solutions

1.1 Pure Algebraic Manipulations Solutions

SOLUTION 1

Simple Linear Equation

Add 7 to both sides of the equation:

$$3x = 11 + 7 \implies 3x = 18$$

Divide both sides by 3:

$$x = \frac{18}{3} = 6$$

SOLUTION 2

Pure Quadratic Equation

Add 18 to both sides:

$$2x^2 = 18 \implies x^2 = 9$$

Take the square root of both sides, accounting for both roots:

$$x = \pm 3$$

SOLUTION 3

Simple Radical Equation

Square both sides of $\sqrt{x+4} = 3$:

$$x + 4 = 3^2 = 9$$

Subtract 4:

$$x = 9 - 4 = 5$$

SOLUTION 4

Simplifying Algebraic Fractions

Factor the numerator using the difference of two squares:

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2}$$

Since $x \neq 2$, cancel the common term $(x-2)$ to get:

$$x+2$$

SOLUTION 5

Factoring Common Terms

Factor out the greatest common factor $3x$:

$$3x^3 - 12x = 3x(x^2-4)$$

Factor the difference of squares:

$$3x(x-2)(x+2)$$

SOLUTION 6

Difference of Squares Mental Trick

Apply the difference of squares identity $a^2 - b^2 = (a-b)(a+b)$ with $a = 105$ and $b = 95$:

$$105^2 - 95^2 = (105-95)(105+95) = (10)(200) = 2000$$

SOLUTION 7

Linear Inequality

Subtract 5 from both sides:

$$-2x > 1 - 5 \implies -2x > -4$$

Divide by -2 and reverse the inequality sign:

$$x < 2 \quad \text{or} \quad (-\infty, 2)$$

SOLUTION 8

Factoring Quadratic Trinomial

Factor the quadratic expression by finding two numbers that multiply to 6 and add to -5 ((-2) and (-3)):

$$(x-2)(x-3)=0$$

Thus, the real solutions are:

$$x=2 \quad \text{and} \quad x=3$$

SOLUTION 9

System of Linear Equations

Given:

$$x+y=10$$
$$2x-y=5$$

Add the two equations to eliminate y:

$$3x=15 \implies x=5$$

Substitute $x=5$ into the first equation:

$$5+y=10 \implies y=5$$

Solution: $(x,y)=(5,5)$.

SOLUTION 10

Variable Substitution

Given $(x-1)^2-4=0$ with $u=x-1$:

$$u^2-4=0 \implies u^2=4 \implies u=\pm 2$$

Since $x=u+1$:

$$x=2+1=3 \quad \text{or} \quad x=-2+1=-1$$

SOLUTION 11

Exponents and Simplification

Express numbers in terms of base 2: $4 = 2^2$ and $8 = 2^3$:

$$\frac{2^5 \cdot (2^2)^2}{2^3} = \frac{2^5 \cdot 2^4}{2^3} = \frac{2^9}{2^3} = 2^{9-3} = 2^6$$

SOLUTION 12

Proportionality Relations

Write the direct proportion equation $y = kx^2$. Substitute $y = 12$ and $x = 2$:

$$12 = k(2^2) \implies 4k = 12 \implies k = 3$$

Now find y when $x = 4$:

$$y = 3(4^2) = 3(16) = 48$$

SOLUTION 13

Clearing Fractional Denominators

Multiply the entire equation $\frac{1}{x} + \frac{1}{2x} = 1$ by $2x$:

$$2 + 1 = 2x \implies 3 = 2x \implies x = \frac{3}{2}$$

SOLUTION 14

Logarithmic Equation

Use the log product rule:

$$\log_{10}(2x) = 1$$

Rewrite in exponential form:

$$2x = 10^1 = 10 \implies x = 5$$

SOLUTION 15

Completing the Square

Given $x^2 + 6x + 2 = 0$. Take half of 6 (3), square it (9), and add/subtract it:

$$(x^2 + 6x + 9) - 9 + 2 = 0 \implies (x+3)^2 - 7 = 0 \implies (x+3)^2 = 7$$

Solve for x:

$$x + 3 = \pm\sqrt{7} \implies x = -3 \pm \sqrt{7}$$

SOLUTION 16

Exponential Quadratic Form

Let $u = 2^x$. The equation $2^{2x} - 5 \cdot 2^x + 4 = 0$ becomes:

$$u^2 - 5u + 4 = 0 \implies (u-1)(u-4) = 0$$

Thus $u = 1$ or $u = 4$. Solving for x:

$$2^x = 1 \implies x = 0 \quad \text{or} \quad 2^x = 4 \implies x = 2$$

SOLUTION 17

Nested Radical Equation

Square both sides of $\sqrt{x+3} = x - 3$:

$$x + 3 = (x-3)^2 \implies x + 3 = x^2 - 6x + 9 \implies x^2 - 7x + 6 = 0$$

Factor the quadratic:

$$(x-1)(x-6) = 0 \implies x = 1 \quad \text{or} \quad x = 6$$

Testing roots: $x = 1$ gives $\sqrt{4} = -2$ (extraneous). Thus, the only valid solution is $x = 6$.

SOLUTION 18

Non-linear System of Equations

Given $xy = 12$ and $x^2 + y^2 = 25$. Use the binomial identity:

$$(x+y)^2 = x^2 + y^2 + 2xy = 25 + 2(12) = 49 \implies x + y = 7 \quad (\text{since } x, y > 0)$$

We have a sum of 7 and product of 12, yielding numbers 3 and 4. Thus solutions are $(3, 4)$ and $(4, 3)$.

SOLUTION 19

Rational Expression Decomposition

Set up partial fractions for $\frac{5x-1}{(x-1)(x+1)}$:

$$\frac{5x-1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} \implies 5x-1 = A(x+1) + B(x-1)$$

Setting $x = 1 \implies 4 = 2A \implies A = 2$. Setting $x = -1 \implies -6 = -2B \implies B = 3$.

$$\frac{2}{x-1} + \frac{3}{x+1}$$

SOLUTION 20

Logarithmic System

Convert $\log_2(x) + \log_2(y) = 3$ to product form:

$$\log_2(xy) = 3 \implies xy = 2^3 = 8$$

Substitute $x = y + 2$ from the second equation:

$$(y+2)y = 8 \implies y^2 + 2y - 8 = 0 \implies (y+4)(y-2) = 0$$

Since $x, y > 0$ implied by logs, $y = 2$, which gives $x = 4$. Solution: $(4, 2)$.

1.2 Physical Algebra Applications Solutions

SOLUTION 21

Constant Speed Relation

Given $d = vt$. If $v' = 2v$ and d is constant:

$$d = (2v)t' \implies t' = \frac{d}{2v} = \frac{t}{2}$$

The time is halved.

SOLUTION 22

Ideal Gas Temperature Isolation

Divide both sides of $PV = nRT$ by nR:

$$T = \frac{PV}{nR}$$

SOLUTION 23

Work Done by a Constant Force

Substitute values into $W = Fd\cos(\theta)$:

$$W = (50\,\text{N})(4\,\text{m})(0.5) = 100\,\text{J}$$

SOLUTION 24

Thermal Heat Capacity

Divide both sides of $Q = mc\Delta T$ by mc:

$$\Delta T = \frac{Q}{mc}$$

SOLUTION 25

Hydrostatic Pressure Depth

Subtract P_0 and divide by ρg:

$$P - P_0 = \rho g h \implies h = \frac{P - P_0}{\rho g}$$

SOLUTION 26

Photon Energy Formula

Substitute $f = \frac{c}{\lambda}$ into $E = hf$:

$$E = \frac{hc}{\lambda}$$

SOLUTION 27

Rotational Energy Velocity

Multiply $K_{\text{rot}} = \frac{1}{2}I\omega^2$ by 2, divide by I, and take the square root:

$$2K_{\text{rot}} = I\omega^2 \implies \omega^2 = \frac{2K_{\text{rot}}}{I} \implies \omega = \sqrt{\frac{2K_{\text{rot}}}{I}}$$

SOLUTION 28

Kinematic Time Isolation

Given $d = \frac{1}{2}at^2$:

$$2d = at^2 \implies t^2 = \frac{2d}{a} \implies t = \sqrt{\frac{2d}{a}}$$

SOLUTION 29

Ohm's Law Power Formula

Substitute $V = IR$ into $P = VI$:

$$P = (IR)I = I^2R$$

SOLUTION 30

Resistors in Parallel

Find a common denominator for the right side of $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$:

$$\frac{1}{R_{\text{eq}}} = \frac{R_2 + R_1}{R_1R_2}$$

Invert both sides:

$$R_{\text{eq}} = \frac{R_1R_2}{R_1 + R_2}$$

SOLUTION 31

Gravitational Inverse-Square Law

Substitute $r' = 2r$ into $F = G\frac{m_1 m_2}{r^2}$:

$$F' = G\frac{m_1 m_2}{(2r)^2} = G\frac{m_1 m_2}{4r^2} = \frac{1}{4}F$$

The force decreases by a factor of 4.

SOLUTION 32

Conservation of Energy Speed

Equate $mgh = \frac{1}{2}mv^2$, cancel m, and solve for v:

$$gh = \frac{1}{2}v^2 \implies v^2 = 2gh \implies v = \sqrt{2gh}$$

SOLUTION 33

Kinematic Distance Formula

Solve $v^2 = v_0^2 + 2ad$ for d:

$$v^2 - v_0^2 = 2ad \implies d = \frac{v^2 - v_0^2}{2a}$$

SOLUTION 34

Spring Potential Energy Stretch

Multiply $U_s = \frac{1}{2}kx^2$ by 2, divide by k, and take the root:

$$2U_s = kx^2 \implies x^2 = \frac{2U_s}{k} \implies x = \sqrt{\frac{2U_s}{k}}$$

SOLUTION 35

Thin Lens Equation Isolation

Subtract $\frac{1}{d_o}$ from $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$:

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{d_o - f}{f\,d_o}$$

Invert both sides to find d_i:

$$d_i = \frac{f\,d_o}{d_o - f}$$

SOLUTION 36

Centripetal Acceleration and Frequency

Substitute $v = 2\pi r f$ into $a = \frac{v^2}{r}$:

$$a = \frac{(2\pi r f)^2}{r} = \frac{4\pi^2 r^2 f^2}{r} = 4\pi^2 r f^2$$

SOLUTION 37

Atwood Machine Acceleration

Add the two equations:

$$T - m_1 g = m_1 a$$
$$m_2 g - T = m_2 a$$

Sum yields:

$$(m_2 - m_1)g = (m_1 + m_2)a \implies a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g$$

SOLUTION 38

Pendulum Period Scaling

Substitute $L' = 4L$ into $T = 2\pi\sqrt{\frac{L}{g}}$:

$$T' = 2\pi\sqrt{\frac{4L}{g}} = 2\pi \cdot 2\sqrt{\frac{L}{g}} = 2T$$

The period doubles.

SOLUTION 39

Simple Binomial Approximation

Rewrite using a negative exponent and apply $(1+x)^n \approx 1+nx$ with $n=-1$:

$$\frac{1}{1+x} = (1+x)^{-1} \approx 1+(-1)x = 1-x$$

SOLUTION 40

Relativistic Kinetic Energy Limit

Express $\gamma = \left(1-\frac{v^2}{c^2}\right)^{-1/2}$. Using the binomial approximation $(1-y)^{-1/2} \approx 1+\frac{1}{2}y$ where $y = v^2/c^2$:

$$\gamma \approx 1+\frac{1}{2}\frac{v^2}{c^2}$$

Substitute into relativistic kinetic energy $K = (\gamma-1)mc^2$:

$$K \approx \left[\left(1+\frac{1}{2}\frac{v^2}{c^2}\right)-1\right]mc^2 = \left(\frac{1}{2}\frac{v^2}{c^2}\right)mc^2 = \frac{1}{2}mv^2$$

This successfully reduces to the classical kinetic energy formula at low speeds ($v \ll c$).